WORKED SOLUTION

STEP III 2009, Question 5

Pure • Symmetric Functions of Roots

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q5

The key idea

Everything here is about the three *elementary symmetric functions*

$$e_1 = x + y + z, \qquad e_2 = yz + zx + xy, \qquad e_3 = xyz,$$

because once we know them, x, y and z are the roots of a known cubic $t^3 - e_1 t^2 + e_2 t - e_3 = 0$. Each new piece of information comes from multiplying two of the given power sums together and seeing which cross terms appear. The recurrence then falls out of the cubic: each of x, y, z satisfies it, so each of their powers does too.

Solution

Write $S_n = x^n + y^n + z^n$, so we are given $S_1 = 1$, $S_2 = 2$, $S_3 = 3$.

The value of $yz + zx + xy$

Squaring S_1,

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(yz + zx + xy),$$

so $1 = 2 + 2(yz + zx + xy)$, giving

$$yz + zx + xy = -\tfrac{1}{2},$$

as required.

The value of $\sum x^2 y$, and then xyz

Write $T = x^2y + x^2z + y^2z + y^2x + z^2x + z^2y$ for the sum of all six terms of the form (one variable squared)$\times$(a different variable). Multiplying S_2 by S_1 gives nine terms: the three cubes and the six terms of T:

$$(x^2 + y^2 + z^2)(x + y + z) = (x^3 + y^3 + z^3) + T.$$

Hence $2 \times 1 = 3 + T$, and so

$$T = -1,$$

which is the second result.

Ask yourself Which product of known quantities contains both T and xyz?

Try $(x+y+z)(yz+zx+xy)$. Multiplying out gives nine terms. Six of them have the shape $x \cdot xy = x^2y$, and these are exactly the six terms of T. The other three are $x \cdot yz$, $y \cdot zx$ and $z \cdot xy$, each equal to xyz. So

$$(x+y+z)(yz+zx+xy) = T + 3xyz.$$

Substituting the values we know, $1 \times (-\frac{1}{2}) = -1 + 3xyz$, so $3xyz = \frac{1}{2}$ and

$$xyz = \tfrac{1}{6},$$

as required.

The recurrence

We now know $e_1 = 1$, $e_2 = -\frac{1}{2}$ and $e_3 = \frac{1}{6}$. Since

$$(t-x)(t-y)(t-z) = t^3 - e_1t^2 + e_2t - e_3 = t^3 - t^2 - \tfrac{1}{2}t - \tfrac{1}{6},$$

each of x, y, z is a root of this cubic. In particular

$$x^3 = x^2 + \tfrac{1}{2}x + \tfrac{1}{6},$$

and similarly for y and z. Multiplying by x^{n-2} gives

$$x^{n+1} = x^n + \tfrac{1}{2}x^{n-1} + \tfrac{1}{6}x^{n-2},$$

and the same with y or z in place of x. Adding the three equations,

$$S_{n+1} = S_n + \tfrac{1}{2}S_{n-1} + \tfrac{1}{6}S_{n-2},$$

so

$$a = 1, \qquad b = \tfrac{1}{2}, \qquad c = \tfrac{1}{6}.$$

Ask yourself Does "for all n" cause any trouble with negative powers?

Multiplying by x^{n-2} is harmless for $n \geq 2$. For smaller n we need negative powers of x, y and z, and these exist because none of them is zero: their product is $\frac{1}{6} \neq 0$. So the relation holds for every integer n. (It holds even if x, y, z are not all real, and in fact they aren't: the cubic has one real root, about 1.431, and two complex conjugate roots.)

Checks

- With $n = 2$ the recurrence should reproduce the given data, using $S_0 = 3$: indeed $S_2 + \frac{1}{2}S_1 + \frac{1}{6}S_0 = 2 + \frac{1}{2} + \frac{1}{2} = 3 = S_3$.
- The recurrence predicts $S_4 = 3 + 1 + \frac{1}{6} = \frac{25}{6}$ and $S_5 = \frac{25}{6} + \frac{3}{2} + \frac{1}{3} = 6$. Solving the cubic numerically (roots 1.43085 and $-0.21542 \pm 0.26471\,\mathrm{i}$) and adding fourth and fifth powers gives 4.16667 and 6.00000, in agreement. The same roots satisfy the recurrence for negative n too.

Common slips

- Getting the signs in the cubic wrong: it is $t^3 - e_1 t^2 + e_2 t - e_3$, so here $b = -e_2 = +\frac{1}{2}$ and $c = e_3 = \frac{1}{6}$.
- Forgetting that $S_0 = 3$, not 1, when testing the recurrence at $n = 2$.

Beyond STEP

These are Newton's identities, which link power sums to elementary symmetric functions for any number of variables. Behind them is the *fundamental theorem of symmetric polynomials*: every symmetric polynomial in $x_1, \ldots, x_k$ is a polynomial in $e_1, \ldots, e_k$. The recurrence for S_n is a linear recurrence whose characteristic equation is the cubic itself, the same idea that solves linear difference equations and gives the Fibonacci formula. Allenby's *Rings, Fields and Groups* develops polynomials and their roots in this spirit.

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